

**Gamifying Self-Care: Using Video Games to Address Motivational and Educational Barriers
in Depression and Anxiety Treatment**

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In the U.S, about 1 in 10 adults have depression. In addition, it is estimated that 5 percent of teens in the U.S. experience moderate to severe depression (*NCCIH Depression*, n.d.). Anxiety disorders are also common among the U.S population with 19.1 percent of adults experiencing them (Harvard Medical School, 2007).

Medication and therapy are widely used treatments for both anxiety and depression, but other practices are often used to fill the gaps. These practices are commonly called “self-care” which was defined in 2021 by Martínez et al. as “the ability to care for oneself through awareness, self-control, and self-reliance in order to achieve, maintain, or promote optimal health and well-being.” (p. 1)

However, the common view of self-care often differs from this definition. Habits such as lying in bed all day or eating junk food are labeled as “self-care” despite their often-harmful nature. Education on proper self-care is important but according to an article from 2024 by the World Health Organization, marginalized populations have poor access to good health information and self-care products can add financial burden. In addition to education barriers, those who are struggling with mental illnesses such as anxiety or depression face more barriers. Difficulty concentrating and fatigue are symptoms of both anxiety and depression according to the DSM-5-TR in 2022. These symptoms can hinder the application of self-care tasks even if those with depression and anxiety are properly educated on effective self-care tasks. I propose the following: the gamification of self-care tasks via video games is an under-explored but promising solution to both the motivational barriers and educational gaps faced by individuals with depression and anxiety.

Before we investigate the efficacy of gamifying self-care tasks via video games, we must determine what self-care tasks are effective. For the purpose of this thesis, self-care tasks will be divided into two main categories, physical and mental tasks. Physical tasks focus on the physical body and include activities such as exercising and eating a nutritious diet. Mental tasks focus on the mind and include activities such as meditation and cognitive reframing.

These next few examples are physical self-care tasks. A study by Dunn et al. from 1998-2001 examined the effect of exercise on depression. Participants aged 20-45 were assigned to one of four aerobic exercise conditions. These varied in frequency (3 days/week or 5 days/week), and energy expenditure (7.0 kcal/kg/week or 17.5 kcal/kg/week). The 17.5 kcal/kg/week condition is the recommended dose of exercise. Energy expenditure had a significant impact on depression at the 12-week mark (depression measured using the 17-item Hamilton Rating Scale for Depression). Exercise frequency had a negligible effect at the 12-week mark. It was found that at least 3 days/week of the 17.5 kcal/kg/week condition had a significant impact (47% decrease from baseline HRSD scores) on depression at the 12-week mark.

A study done by Zheng et al. in 2024 looked at the relationship between exercise and mental health in college students with a sample size of 2891. Lifestyle, demographic and psychological (anxiety and depression) data were collected. Depression was reported by 44.2% of the students and anxiety was reported by 39.5% of the students. The more frequent the exercise, the less anxiety and depression symptoms were experienced. Higher sports frequency and longer sports time were associated with lower rates of depression and anxiety. Sports such as basketball, dancing, running, and tennis were associated with lower levels of depression and

anxiety. Activities such as prolonged sedentary times, stressors, and increased electronic device usage time were associated with higher depression scores. A preference for eating vegetables was associated with lower depression scores. Scores of depression and anxiety were lowest among the 5/week sports group, but it was not profoundly different from the 3-4/week sports group.

In a study done by Francis et al. in 2019 participants had moderate or greater depression, had a poor diet and were ages 17-35. The diet included 5 servings of vegetables/day, 2-3 servings of fruits/day, 3 of whole-grain cereals/day, 3 of protein/day, 3 of unsweetened dairy/day, 3 tablespoons of nuts/day, 3 tablespoons of olive oil/day, 1 teaspoon of spices (cinnamon/turmeric)/day, and 3 of fish/week. Participants were also instructed to reduce their intake of sugars, refined carbs, fatty/processed meats and soda. A significant drop in rates of depression (measured with the Centre for Epidemiological Studies Depression Scale; CESD-R) was found. Improvement was also seen in the stress and anxiety subscales of the DASS-21.

This next example would be an example of a mental self-care task. A study done by Lerma et al. in 2017 looked at the effects of Cognitive Behavioral Therapy (CBT) on the depression and anxiety symptoms of chronic hemodialysis patients. The study found a decrease in depression symptoms (-54%), anxiety symptoms and cognitive distortion. Cognitive restructuring is a technique used in CBT that involves identifying automatic negative thoughts and challenging them by using a series of questions. (Rohan, 2008)

Self-care tasks such as exercising or eating well are beneficial but can be difficult to complete consistently. This is where gamification comes in. Merriam Webster Dictionary defines gamification as “the process of adding games or game like elements to something (such as a

task) so as to encourage participation.” A literature review done by Johnson et al. in 2016 found that gamification can be intrinsically motivating and can support wellbeing. Johnson et al. also found that using digital technology in gamification has benefits such as broad accessibility, broad appeal, broad applicability, cost-benefit efficiency. Already, gamification has seen great success in educational software, such as Duolingo, a popular app used to learn languages.

One way that digital technology can be used to gamify self-care tasks is through video games. According to the Entertainment Software Association’s annual “Essential Facts About the U.S. Video Game Industry” report in 2025, 60% of adults in the U. S. play video games every week. The World Health Organization says in their article from 2024 “Self-care for health and wellbeing” that we need “...innovative strategies that go beyond a conventional health sector...” to “...address these challenges in accessing quality healthcare.” Using video games to gamify self-care tasks can reach a wide audience, which can help educate many people on effective self-care tasks. Already, there are well-known games and apps that use gamification to help motivate users to complete tasks. Just Dance, a popular game franchise by Ubisoft, allows users to learn dances and perform the moves in real time alongside digital dance instructors. Pokémon GO, a mobile app by Nintendo, encourages players to walk around the real world to collect mystical animals and battle against other players. These pieces of technology focus on keeping the user healthy by gamifying exercise. A crucial component missing from many of these games is the mental component. Exercise and dietary self-care tasks are more frequently gamified than self-care tasks like cognitive restructuring.

A meta-analysis done by Lau et al. in 2017 found that serious gaming interventions may help reduce disorder-related symptoms. Two studies (Flemming et al. and Merry et al.) analyzed by Lau et al. looked at the effects of a game, SPARX, on depression.

A systematic review done by Primack et al. in 2012 found that in the 38 studies examined, video games improved 69% of psychological therapy outcomes, 59% of physical therapy outcomes, 50% of physical activity outcomes, and 42% of health education outcomes. They also found that video games that had either a physical or psychological therapy aspect were more successful at achieving their goals.

There are many ways to gamify self-care tasks. Apostol et al. outline 8 main features of a game in their paper from 2013: rules, clear goals, feedback and rewards, problem solving, story, player(s), a safe environment, and a sense of mastery. These features can be combined with self-care tasks to gamify the tasks. As discussed earlier, physical exercise, eating a diet with more fruits and vegetables, and CBT can all help improve the symptoms of depression and/or anxiety so these are the tasks that will be used as examples of gamification. Physical exercise can be gamified through walking and the use of step counters. By quantifying physical exercise using a step counter, the user gains the clear goal of meeting a certain number of steps. This clear goal can then be incorporated into a story using a non-player character (NPC). An NPC could ask a user to fetch them an item that is far away, and the user must then take a certain number of steps to get the item for the character.

The consumption of fruits and vegetables could be gamified by counting the number of servings of fruits and vegetables the user consumes. Like the previous task, this gives the user a

clear goal. To gamify the task even further, the user could be rewarded with new plants in their garden based on the servings of fruits and vegetables they consume.

CBT can be gamified using cognitive restructuring. As mentioned before, cognitive restructuring helps the user challenge negative thoughts through a series of questions about their negative thought. This could be gamified by using a character in a game to ask the questions. When a user finishes the questions for a thought that thought could be counted as completed. The user could be rewarded with decorations for their home for each thought challenged.

Depression and anxiety are both significant problems among the U.S. population. Medication and therapy are common treatments, but self-care tasks such as exercise, eating produce, and cognitive restructuring, can be effective treatments as well. Self-care tasks can be combined with gaming elements to help users complete self-care tasks more effectively. Adding clear goals, rewards, and a story can gamify self-care tasks. The gamification of self-care tasks via video games should be explored further as it shows potential for the treatment of anxiety and depression.

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